

PRACTICAL EXAM TRAINER GUIDE

WATER & PREVENTIVE MAINTENANCE FOUNDATION

COFFEE TECHNICIANS PROGRAM

实作测验 讲师指南

水&预防性维护基础课程

咖啡技师学程

This practical exam simulates tasks which are commonly performed during a preventive maintenance visit. These tasks are divided into four (4) Sections. To receive full credit for each section the learner must record all relevant observations, calculations, and task completions requested in the Practical Exam Learner Worksheet.

本实务测验模拟在预防性维护检测时常见的项目。这些项目分为四个部分。要获得每一个部分的学分，学员必须记录下实务测验学员工作表中所要求的所有相关观察、计算和任务完成情况。

IMPORTANT SAFETY NOTICE: 重要安全事项：

This exam includes activities which involve the learner interacting with hand tools. It is the sole responsibility of the AST to ensure that:

本测验包含让学员操作手动工具的环节。SCA 讲师需负责确保下列事项：

All Learners are properly supervised at all times while working with hand tools

- 所有学员随时都在适当的监督下操作手动工具。

An easily accessible, fully-equipped first aid kit is on site

- 现场备有完整配备且方便取用的急救箱。

All Learners use appropriate Personal Protective Equipment during course activities

- 所有学员在课程活动期间皆使用适当的个人防护配备。

All working stations are set up to meet the safe workplace recommendations detailed in guidebook sections 1.01 and 1.06 and all local workplace safety standards

- 所有工作站的设置皆须达到指引手册章节 1.01、1.06 所载明之安全建议及所有当地工作场域安全标准。

If unsafe behavior is observed at any time, the exam is stopped for the Learner and a score of zero (0) is to be given for that section of the exam

- 无论何时，当注意到学员做出危险举动时，该学员的测验立即中止，并于该部分的测验给予(0)分。

Learners exhibiting unsafe behavior are provided with feedback and demonstrate an understanding of safe behavior before attempting a retake

- 须对学员的危险举动给予回馈，并在尝试重考前确认学员对安全行为的理解。

Learners are familiar with the safe use of all hand and measurement tools to be used prior to starting this exam

- 在开始本测验前，学员需要熟悉所有手动工具及测量工具的安全使用方式。

All course activities are conducted in accordance with local safety regulations and best practices

- 所有课程活动皆按照当地安全规定及最佳实务进行。

All stations which involve live electricity are connected in accordance with local electrical code and workplace safety regulations

- 所有需用电之工作站皆须依照当地电工标准及工作场域安全规定进行连接。

All stations which involve live electricity are connected to power using GFCI devices which have been tested for proper function before beginning this exam.

- 所有需用电之工作站连接到电源皆须使用漏电断路设备，并在测验前通过功能测试。

All outlets and devices are properly grounded and tested by the AST to confirm proper grounding before beginning this exam

- 所有插座及设备在测验前皆确实接地并经由讲师确认。

All hand tools and measurement devices are inspected for safe and proper function before beginning this exam

- 所有手动工具及测量设备在测验前皆经测试后确认安全性及功能正常。

If local electrical code or workplace safety regulations do not allow for a prescribed activity, the AST is not to conduct this activity and is to contact SCA Education to discuss alternative activities.

假如当地安全电工标准或工作场域安全规定不允许进行指定的活动，讲师将不进行该活动并联络SCA 教育团队讨论替代方案。

Section 1 – Measuring Water Composition

第一章 - 测量水的组成

- **Time Available: 10 minutes for all three (3) parts**
- **Points Available: 15, 11 points required to pass**
- **Silent working is required.**
- **This section is designed to evaluate the Learner's ability to:**
 - **Measure the Total Hardness and Alkalinity of a water sample**
 - **Record Total Hardness and Alkalinity measurements on the SCA Water Chart**
 - **Use an Electrical Conductivity meter to measure the estimated TDS of a water sample**

测验时间：总共十分钟，分三个部分

得分：15分，通过门坎为11分

作业中禁止交谈

本章节旨在评估学员下列能力：

测量水质样本的总硬度及碱度。

将总硬度及碱度的测量结果记录在SCA水质图上。

使用导电度计测量水质样本的TDS概略值。

Station Preparation 工作站前置作业

Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms.

- 利用在学员间设置视线挡板或是分配至独立工作间以隔离出本测验的个人工作空间。准备两个大
- Prepare 2 different approximately 500 mL samples of water for evaluation by the learners:

- 约 500ml 的不同水质样本供学员们评估

Label each sample with a 3-digit number

- 以 3 位数字为各个水质样本做样本编号

Samples must have the following parameters:

- 水质样本必须符合下列参数：

- Total Hardness between 30 and 160 ppm CaCO₃
- Alkalinity between 30 and 160 ppm CaCO₃
- Samples must differ from each other on both parameters by at least 20 ppm

CaCO₃ 总硬度介于 30~160ppm CaCO₃ 之间

碱度介于 30~160ppm CaCO₃ 之间

水质样本之间在两个参数上皆须有 20ppm CaCO₃ 的差异

- Samples may be prepared using prepared coffee mineral mixes, AST-mixed concentrate solutions, or bottled water. See The SCA Water Quality Handbook section __ for guidance

水质样本的备制可使用调制好的咖啡矿物质调和物、AST 调和浓缩液或是瓶装矿泉水。

可参阅 SCA 水质手册第 _ 章节寻求指引。

Provide drop-count Total Hardness and Alkalinity test kits, electrical conductivity TDS meters, and necessary sampling and testing accessories and supplies

- 提供测验总硬度及碱度的滴定试剂、导电度 TDS 计、必要的采样及测试器材。

Provide printed copies of the SCA Water Chart, 1 per student minimum

- 提供 SCA 水质图的复印件，至少每位学生一份。

Provide a visual reference for the student to use during this section of the exam which indicates the color change expected for each drop-count test and the conversion factor between drops required for color change and ppm CaCO₃

- 提供标示出各个滴定试验所预计的颜色变化，以及颜色变化所需的滴定量和 ppm CaCO₃ 的换算系数之视觉参照，供学生于本章节测验中使用。

Provide each Learner with a calculator

- 提供各个学员一台计算器。

Provide each Learner with a clipboard.

- 提供各个学员一块书写夹板。

Trainer Instructions 讲师指示

Review the Learner Instructions of the exercise with the Learner.

- 陪同学员详读学员测验指示

Assign a water sample to the Learner for evaluation. A sample may be evaluated by more than one learner. If a retake is required for this section, assign a different sample to the Learner for their second attempt.

- 分配水质样本供学员评估。一个水质样本可能由多个学员进行评估。如果本章节需要重考，请在第二次尝试时为学员分配一个不同的水质样本。

If the Learner is unable to obtain Total Hardness and Alkalinity measurements in the Skill A portion of this section, provide values for Total Hardness and Alkalinity within the SCA Core Zone for their use in Skill B.

- 如果学员在本章节的技能 A 部分，无法获得总硬度及碱度测量值，请提供位于 SCA 水质核心区的一组总硬度及碱度数值予学员们用于技能 B 部分的测验。

Alert the Learner to begin and start the timer.

- 提醒学员测验开始并开始计时。

The Learner is to alert you when they have finished.

- 学员将会在完成测验时告知你。

Once the Learner has announced that they have finished, they cannot add to or change their answers. If time expires, the Learner cannot add to or change their answers.

- 一旦学员宣布他们完成测验时，他们将无法增加或修改答案。如果时间结束时，学员亦无法增加或修改答案。

Silent work is required. If a Learner compromises the integrity of the exam for another, you must reset the exam for the next Learner.

- 作业中禁止交谈。如果有学员破坏了别的学员考试之完整性，请为下一位学员重新设置测验。

Learner Instructions 学员指示

A designated water sample has been prepared for your use in this section of the exam. You are to measure several parameters of this sample and record your results.

- 指定的水质样本已准备好以供本章节测试中使用。你将需要测量该水质样本的几个参数并记录结果。

There are three (3) parts to this section of the exam.

- 本章节测验一共有 3 个部分。

Read the instructions for each part thoroughly.

- 请详阅各个部份的说明。

The trainer will tell you when your time begins.

- 计时开始时，讲师将会告知你。

Alert your trainer when you have completed this section.

- 完成本章节测验时请通知你的讲师。

Once you have alerted the trainer, or if time expires, you will not be able to change your answers.

- 一旦你通知讲师或是时间结束，你将无法再修改答案。

Skill A - Measure the Total Hardness and Alkalinity of a water sample

技能 A - 测量水质样本的总硬度及碱度

Use the provided test kit to measure the Total Hardness and Alkalinity of the provided sample. You may reference the visual aid which the AST has provided.

- 使用提供的试剂测量水质样本的总硬度及碱度。你可以参考讲师所提供的视觉参照。

Record your result on the Sample Preventive Maintenance Checklist.

- 请将你的结果记录在预防性维护核对表。

Skill B - Record Total Hardness and Alkalinity measurements on the SCA Water Chart

技能 B - 将总硬度及碱度测量结果记录在 SCA 水质图

A copy of the SCA Water Chart has been provided for your use in this section.

- 本章节提供了一份 SCA 水质图复印件供你使用。

Perform any calculations which may be required to convert the measurements you obtained in Skill A of this section into ppm CaCO₃. You may reference the visual aid which the AST has provided.

- 将你在本章节技能 A 所得之测量值，执行所需的运算转换成 ppm CaCO₃ 的值。你可以参考讲师提供的视觉参照。

If you were unable to obtain measurements in Skill A, your AST will provide values for your use.

- 如果你没有在技能 A 取得测量值，讲师将会提供数值予你使用。

Plot the Total Hardness and Alkalinity values you obtain on this chart.

- 在水质图上标示你获得的总硬度及碱度值。

Skill C - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

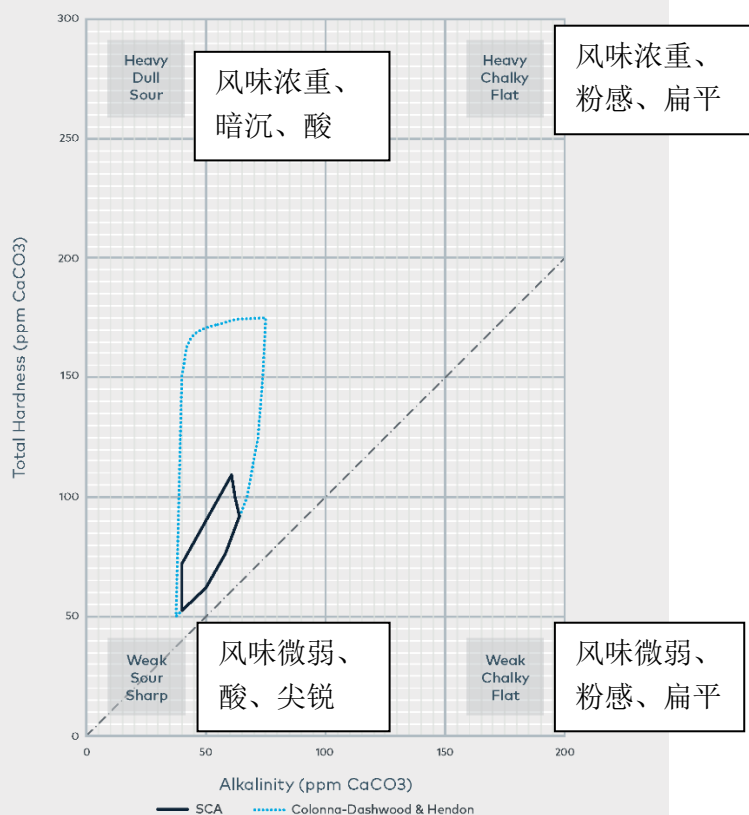
技能 C - 使用导电度计测量水质样本的 TDS 概略值。

Use the provided Electrical Conductivity meter to measure the estimated TDS of the provided water sample.

- 使用提供的导电度计测量水质样本的 TDS 概略值。

Record your results on the Sample Preventive Maintenance Checklist

- 请将你的结果记录在预防性维护核对表。



Scoring 计分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

将得出的小计总分填入实务测验学员工作表的对应表格

Skill 1A - Measure the Total Hardness and Alkalinity of a water sample

技能 1A-测量水质样本的总硬度及碱度

Task 任务	Points Possible 可能得分
Learner properly collects samples and fills testing vials 学员正确地收集水质样本并装满试瓶。	1
Learner uses appropriate technique for adding reagent drops and agitating sample 学员使用适当的技巧添加滴定试剂并搅拌水质样本。	1 per vial 每瓶 1 分
Learner correctly recognizes color change and drops added for Total Hardness 学员正确地辨认测量总硬度时的颜色变化及添加的滴定量。	1
Learner correctly recognizes color change and drops added for Alkalinity 学员正确地辨认测量碱度时的颜色变化及添加的滴定量。	1
Learner correctly records measurements on Learner Worksheet 学员正确地将测量结果记录在学员工作表。	1
Total 总分	6

Skill 1B - Record Total Hardness and Alkalinity measurements on the SCA Water Chart

技能 1B-在 SCA 水质图记录总硬度及碱度测量值

Task 任务	Points Possible 可能得分
Learner correctly performs calculation to convert measurements into ppm CaCO ₃ 学员正确地执行运算将测量结果转换成 ppm CaCO ₃ 值。	1 per value 每个数值 1 分
Learner accurately plots the Total Hardness result obtained on the provided chart 学员准确地在提供的图表中标示出所获得的总硬度结果。	2
Learner accurately plots the Alkalinity result obtained on the provided chart 学员准确地在提供的图表中标示出所获得的碱度结果。	2

	Total 总分	6
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Skill 1C - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

技能 1C-使用导电度计测量水质样本的 TDS 概略值。

Task 任务	Points Possible 可能得分	
Learner properly collects water sample 学员正确地收集水质样本。	1	
Learner uses appropriate technique for using electrical conductivity meter 学员使用适当的技巧操作导电度计。	1	
Learner correctly records estimated TDS measurement on Learner Worksheet 学员正确地将 TDS 测量结果记录在学员工作表。	1	
Total 总分	3	

Section 2 - Water Treatment System Maintenance

第二章-水处理系统维护

- **Total Time Available: 10 minutes**
- **Points Available: 13, 9 points required to pass**
- **Silent working is required.**
- **This section is designed to evaluate the Learner's ability to:**
 - **replace a cartridge in a drop-in style water filter system**

测验时间：共十分钟

得分：13分，通过门槛为9分

作业中禁止交谈

本章节旨在评估学员下列能力：

更换置入式滤水器系统中的滤芯。

Station Preparation 工作站前置准备

- Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms.
利用在学员间设置视线挡板或是分配至独立工作间以隔离出本测验的个人工作空间。
- Prepare a drop-in style water filter system and connect to water supply:
准备一组置入式滤水器系统并连接至供水。
 - System must be connected to either a building's plumbing or bottled water pumps using a readily accessible shut-off valve.
系统必须连接到自来水或是瓶装水泵浦，使用方便使用的止水阀。
 - Filter system should be securely mounted in reasonable proximity to the water supply.
过滤系统须被牢固地安装在合理靠近供水的地方。
 - A cartridge of any type must be installed in the filter housing
每个类型的滤芯皆须被安装在滤水器外壳中。
 - Provide one "new" filter cartridge for each Learner. The cartridge need not be unused but must be dry.
为每位学员提供一组 " 新的 " 滤芯。该滤芯不需要是未被使用过的，但必须是干燥的。
 - The filter system outlet must be connected to a shut-off valve for use in flushing the filter after replacement. This connection may utilize a tee if the filter system is also connected to a piece of coffee equipment.
过滤系统出水口须被连接到止水阀，在更换滤芯后用于冲洗滤水器。假如过滤系统还连接到别台咖啡设备，该接头可以使用三通管
- Provide a filter housing wrench, container to capture water during the flushing process, and all materials required for replacing the filter cartridge.
提供一只滤水器外壳扳手，冲洗过程中用于盛水的容器，以及所有更换滤芯所需的材料。

Trainer Instructions 讲师指示

- Review the Learner Instructions of the exercise with the Learner.
陪同学员详读学员测验指示

- **Alert the Learner to begin and start the timer.**
通知学员测验开始并开始计时
- Observe and evaluate their work and score the task according to the table below.
观察并评估他们的作业情况并根据下列表格对任务进行评分。
- **Alert the Learner to begin and start the timer.**
通知学员测验开始并开始计时
- The Learner is to alert you when they have finished.
学员将会在完成时告知你。
- If time expires prior to the completion of the task, the learner may receive partial credit for the work that has been completed.
假如时间在完成任务前结束，学员可以从已经完成的作业中得到部份分数。

Learner Instructions 学员指示

- A drop-in cartridge style water filtration system which is connected to a water supply has been prepared for your use in this section of the exam.
在本章节测验中，已准备一组连结上供水的置入式滤水器系统供你使用。
- Service records for this filter system indicate that this cartridge was installed one year prior to today's date. Record this date in the appropriate space on your Learner Worksheet.
本过滤系统的维护纪录显示该滤芯是在今日日期的一年前安装的。将该日期记录在学员工作表适当的位置中。
- The trainer will tell you when your time begins.
你的测验时间开始时讲师将会告知你。
- You are to perform all necessary steps to replace the filter cartridge and return the system to service.
执行全部所需的步骤，完成滤芯的更换并让系统恢复运作。
- When you have completed this section, mark the task as completed on your Learner Worksheet and alert your trainer.
当你完成本章节时，在你的学员工作表上注记该任务已完成并通知讲师。

Scoring 计分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

将得出的小计总分填入实务测验学员工作表的对应表格

Replace a cartridge in a drop-in style water filter system

更换置入式过滤系统的滤芯

Task 任务	Points Possible 可能得分
Learner turns off water supply to water filter 学员关闭滤水器的供水。	2
Learner removes filter housing and old water filter cartridge 学员拆下滤水器外壳并移除老旧滤芯。	2

Learner properly cleans and sanitizes filter housing 学员正确地清洁并消毒滤水器外壳。	2
Learner installs new water filter cartridge and reassembles housing properly 学员安装新的滤芯并正确地重新组装外壳。	2
Learner turns on water supply to filter and flushes system appropriately 学员打开滤水器的供水并适当地冲洗过滤系统。	2
Filter system does not leak 过滤系统不会漏水。	2
Learner records last change date and replacement on Learner Worksheet 学员在学员工作表上记录最后更换日期及更换项目。	1
Total 总分	13

Section 3 - Coffee Brewer and Espresso Grinder Preventive Maintenance

第三章-咖啡冲煮机及义式磨豆机预防性维护

- Total Time Available: 20 minutes for both (2) parts.
- Points Available: 29, 20 points required to pass
- Silent working is required.
- This section is designed to evaluate the Learner's ability to:
 - Inspect and clean a **coffee batch brewer** level probe
 - Replace the burrs in an espresso grinder

测验时间:两个部份各 20 分钟

得分:29 分, 通过门槛为 20 分

作业中禁止交谈

本章节旨在评估学员下列能力:

检查并清洁滴滤式咖啡机的水位探针。

更换义式磨豆机的刀盘。

Station Preparation

工作站前置准备

- Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms. One learner should not be able to see the work of a fellow learner.
利用在学员间设置视线挡板或是分配至独立工作间以隔离出本测验的个人工作空间。
学员不应能够看见邻近学员的作业情况。

For Skill A

在技能 A

- Prepare a coffee batch brewer with a removable water level probe.
准备一台配有可拆卸水位探针的滴滤式咖啡机
 - This brewer **must not** be connected to electrical power
该咖啡机不得接上电源。
 - The plug end of the power cord must be visible for the Learner to verify that it is unplugged.
电源线的插头端必须在学员的视线内以确保未被插上。
 - The brewer may, at the trainer's discretion, be connected to a water supply and the tank may be full of water.
根据讲师的判断, 该咖啡机可以接上供水并在水箱装满水。
- Provide all tools and supplies needed to remove and clean level probe, possibly including:
提供拆卸及清洁水位探针所需的所有器材, 可能包含:
 - Screwdriver
 - Wrench
 - Pliers
 - Utility knife
 - Steel wool or scrubbing pad
螺丝起子
扳手

钳子
工具刀
钢丝绒或菜瓜布

For Skill B

在技能 B

- Prepare a fully-operational espresso grinder and matching electrical outlet.
准备一台可完整运作的义式磨豆机及合适的电源插座
 - This grinder **must not** be connected to electrical power
该磨豆机不得接上电源
 - The plug and outlet must be accessible so the Learner may plug it in to confirm function after completing the test activities.
插头及插座必须方便取用，让学员能在完成测验项目后插上以确认功能。
- Provide all tools and supplies needed to access and remove grinding burrs, possibly including:
提供检查及拆除刀盘所需的所有器材，可能包含：
 - Screwdrivers
螺丝起子
 - Wrench
扳手
 - Stiff-bristled brush
硬毛刷
 - Steel wool or scrubbing pad
钢丝绒或菜瓜布
 - Appropriate lubricant
合适的润滑剂

Trainer Instructions 讲师指示

- Review the Learner Instructions of the exercise with the Learner.
陪同学员详读学员测验指示。
- Alert the Learner to begin and start the timer.
通知学员测验开始并开始计时。
- The learner will alert you when they have reached certain stages of the task. Evaluate their work and score the task according to the table below. Stop the timer while work is being evaluated.
学员每完成部份阶段的任务时将会通知你。评估他们的作业情形并根据下表进行评分。
作业评量中暂停计时。
- If time expires prior to the completion of the task, the learner may receive partial credit for the work that has been completed.
如果时间在任务完成前结束，学员可以从已完成的作业中得到部份分数。
- To receive full credit, the learner must perform the tasks correctly and mark all tasks correctly in the Learner Practical Exam Worksheet
要获得满分，学员必须正确地执行任务并在实务测验学员工作表标记出来

- This section includes working with tools. The learner must demonstrate safe working practices at all times during this section. Any unsafe actions will result in a score of 0 points for this section.
本章节包含工具操作。在测验中学员必须随时展现出安全的工作实践。出现任何不安全的行为，本章节将会被评为 0 分。

Learner Instructions 学员指示

- There are two (2) parts to this section of the exam.
本章节测验包含(2)个部分。
- Read the instructions for each part thoroughly.
请详阅每个部份的说明。
- The trainer has provided a coffee batch brewer and fully-operational espresso grinder.
讲师已提供一台滴滤式咖啡机及能完整运作的义式磨豆机。
- Service records for this equipment indicate that they were last serviced one year prior to today's date.
该设备的维护纪录显示最后维护日期为今日起算一年以前。
- In this section, you are to demonstrate a series of skills, detailed below, to complete preventive maintenance service on the provided batch coffee brewer and espresso grinder.
在本章节中，你需要展示下文所述的一系列技能，来完成对提供的滴滤式咖啡机及义式磨豆机的预防性维护服务。
- Record each task as complete on the Learner Practical Exam Worksheet, including observations where required.
在学员实务测验工作表上将各个任务记录为完成，并记录下要求的各项观测结果。
- The trainer will tell you when your time begins.
当你的测验时间开始时，讲师将会告知你。
- Alert your trainer when you have completed this section.
当你完成这个本章节时请通知讲师。
- **This section includes working with tools. Safe working practices must be demonstrated at all times during this section. Any unsafe actions will result in a score of 0 points for this section.**
本章节包含工具操作。在测验中必须随时展现出安全的工作实践。
任何不安全的行为将造成本章节被评为 0 分。

Skill A - Inspect and clean a coffee batch brewer level probe

技能 A-检查并清洁滴滤式咖啡机的水位探针。

- The necessary tools are at this station.
所需的工具备妥在工作站中。
- Use the provided tools to:
使用提供的工具完成下列事项:
 - Remove any machine panels required to access the level probe.
拆除任何所需的机械面板以便检查水位探针。
 - Remove the batch brewer level probe.
拆除滴滤式咖啡机的水位探针。
 - Inspect the probe for scale build-up and condition.
检查探针上的结垢及探针的状况。
 - Clean to remove any scale build-up
进行清洁以去除任何结垢。

- Reinstall level probe.
重新安装水位探针。
- Once the level probe has been reinstalled, alert the trainer to inspect your work, then:
当水位探针已被重新安装，请通知讲师来检查你的作业，然后：
 - Replace all machine panels.
重新安装所有机械面板。

Skill B - Replace the burrs in an espresso grinder

技能 B-更换义式磨豆机的刀盘

- The necessary tools and replacement parts are at this station.
所需的工具及更换零件备妥在工作站中。
- Use the provided tools to
使用提供的工具完成下列事项：
 - Disassemble the grinder as needed to access the grinding burrs.
根据需求拆开磨豆机以检查刀盘。
 - Remove the current set of grinding burrs, noting condition.
拆除目前的刀盘组，注意拆除的状况。
Clean the grinding chamber and burr carriers.
清洁研磨腔室及刀座。
- Once the grinding chamber and burr carriers are sufficiently clean, alert the trainer to inspect your work, then:
当研磨腔室及刀座清洁干净时，请通知讲师检查你的作业，然后：
 - Install replacement grinding burrs.
安装替换的刀盘。
 - Reassemble the grinder.
重新组装磨豆机。
 - Connect the grinder to electrical power and test for proper function.
将磨豆机接上电源并测试功能是否正常。
 - Adjust to the approximately-correct setting
调整至大略正确的设定。

Scoring 计分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

将得出的小计总分填入实务测验学员工作表的对应表格

Skill 3A - Clean a coffee batch brewer level probe

技能 3A- 清洁滴滤式咖啡机的水位探针

Task 任务	Points Possible 可能得分
Learner confirms that coffee brewer is disconnected from electrical power	2

学员确认滴滤式咖啡机未接上电源。	
Learner removes required access panel and places screws in a secure location 学员拆除所需的检修孔盖并将螺丝放置在妥善的位置。	2
Learner removes level probe without damaging probe or mounting bushings 学员在不损害探针或安装套管的情况下拆除水位探针。	2
Learner uses appropriate means to fully clean the level probe 学员使用适当的方法完整地清洁水位探针。	2
Learner reinstalls level probe and connects wires if required 学员重新安装水位探针并根据需求接上电线。	2
Learner reassembles access panel and tightens screws appropriately 学员重新安装检修孔盖并适当地拧紧螺丝。	2
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1
Total 总分	13

Skill 3B - Replace the burrs in an espresso grinder

技能 3B-更换义式磨豆机的刀盘

Task 任务	可能得分 Points Possible
Learner unplugs espresso grinder before beginning service 学员在开始维护前拔除义式磨豆机的插头。	2
Learner opens grinding chamber and removes worn grinding burrs 学员拆开研磨腔室并移除磨损的刀盘。	3
Learner thoroughly cleans grinding chamber and burr mounting surfaces 学员彻底地清洁研磨腔室及刀盘的安装面。	2
Learner installs new grinder burrs, using an appropriate screw tightening procedure 学员使用合适的螺丝拧紧方式安装新的刀盘。	3
Learner reassembles grinder and connects to electrical outlet	3

学员重新安装磨豆机并接上插座。	
Learner tests for proper function and adjusts to approximately correct setting 学员测试功能是否正常并调整至大略正确的设定。	2
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1
Total 总分	16

Section 4 - Espresso Machine Preventive Maintenance

第四章-义式咖啡机的预防性维护

- **Total Time Available: 20 minutes for all three (3) parts.**
- **Points Available: 43, 30 points required to pass**
- **Silent working is required.**
- **This section is designed to evaluate the Learner's ability to:**
 - replace an espresso machine group screen and gasket
 - rebuild an espresso machine steam wand joint
 - record espresso machine service activity and inspection observations on a preventive maintenance checklist

测验时间:总共 20 分钟,分三个部份

得分:43 分,通过门坎为 30 分

作业中禁止交谈

本章节旨在评估学员下列能力:

更换义式咖啡机的冲煮头分水网及垫圈。

重新组装义式咖啡机的蒸气棒接头。

在预防性维护核对表上记录义式咖啡机维护活动及检查观测结果

Station Preparation 工作站前置准备

- Prepare a fully-operational espresso machine which **is connected to service**, powered on, and adjusted to manufacturer specifications.
准备一台能完整运作的义式咖啡机,并**接上服务**、电源且调整至符合制造商规范的设定。
- The espresso machine must utilize group screens which are retained by a screw.
义式咖啡机必须使用由螺丝固定的冲煮头分水网。
- The espresso machine may utilize a ball style of steam wand joint or an alternative configuration.
义式咖啡机可以使用球形蒸气棒接头或替代配置。
- Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms. One learner should not be able to see the work of a fellow learner.
利用在学员间设置视线挡板或是分配至独立工作间以隔离出本测验的个人工作空间。学员不应能够看见邻近学员的作业情况。

For Skill A

在技能 A

- Provide all necessary tools and supplies, including:
提供所有需要的器材,包含:
 - Group gasket
 - Group screen
 - Group screen screw
 - Screwdriver
 - Pick awl
 - Stiff-bristled brush冲煮头垫圈

冲煮头分水网
冲煮头分水网螺丝
螺丝起子
尖锥
硬毛刷

For Skill B

在技能 B

- Provide all necessary tools and supplies, including:
提供所有需要的器材，包含：
 - Steam wand joint wear parts as-recommended by the espresso machine manufacturer
该义式咖啡机制造商建议的蒸气棒接头耗材
 - Appropriate lubricant
 - Appropriately-sized wrench(es), if required
 - Snap ring pliers, if required
 - Pick awl
 - Stiff-bristled brush
 - Steel wool or scrubbing pad
合适的润滑剂
合适尺寸的扳手(如果有需求)
卡环钳(如果有需求)
尖锥
硬毛刷
钢丝绒或菜瓜布

For Skill C

在技能 C

- Provide a blank portafilter basket or thermofilter device for pump pressure observation
提供一个无孔滤杯或是测温测压把手以供观察泵浦压力。

Trainer Instructions 讲师指示

- Review the Learner Instructions of the exercise with the Learner.
陪同学员详读学员测验指示。
- Designate one brewing group and one steam wand joint for each Learner to service.
为各个学员指定一个冲煮头及一只蒸气棒接头进行维护。
- Alert the Learner to begin and start the timer.
通知学员测验开始并开始计时。
- If time expires prior to the completion of the task, the learner may receive partial credit for the work that has been completed.
假如时间在完成任务前结束，学员可以从已经完成的作业中得到部份分数。
- To receive full credit, the learner must perform the tasks correctly and mark all tasks correctly in the Learner Practical Exam Worksheet

为了得到满分，学员必须正确执行任务并在学员实务测验工作表上正确地标记所有任务

- This section includes working with tools. The learner must demonstrate safe working practices at all times during this section. Any unsafe actions will result in a score of 0 points for this section.

本章节包含工具操作。在测验中学员必须随时展现出安全的工作实践。出现任何不安全的行为，本章节将会被评为 0 分。

Learner Instructions 学员指示

- There are three (3) parts to this section of the exam.
本章节测验包含(3)个部分。
- Read the instructions for each part thoroughly.
请详阅每个部份的说明。
- The trainer has provided a fully-operational espresso machine.
讲师提供一台能完整运作的义式咖啡机。
- Service records for this machine indicate that it was last serviced one year prior to today's date
该设备的维护纪录显示最后维护日期为今日起算一年以前。
- In this section, you are to demonstrate a series of skills, detailed below, to complete preventive maintenance service on the provided espresso machine, including: a thorough inspection, servicing one brewing group, and servicing one steam wand joint.
在本章节中，你需要展示下文所述的一系列技能，来完成对提供的义式咖啡机的预防性维护服务，包含：一次彻底的检查、维护一个冲煮头以及维护一只蒸气棒接头。
- Record each task as complete on the Learner Practical Exam Worksheet, including observed values where required.
在学员实务测验工作表上将各个任务记录为完成，并记录下要求的各项观测到的数值。
- The trainer will tell you when your time begins.
当你的测验时间开始时，讲师将会告知你。
- Alert your trainer when you have completed this section.
当你完成这个本章节时请通知讲师。
- **This section includes working with tools. Safe working practices must be demonstrated at all times during this section. Any unsafe actions will result in a score of 0 points for this section.**
本章节包含工具操作。在测验中必须随时展现出安全的工作实践。任何不安全的行为将造成本章节被评为 0 分。

Skill A - Replace an espresso machine group screen and gasket

技能A-更换一组义式咖啡机的冲煮头分水网及垫圈

- The trainer has indicated which group is to be serviced
讲师已指定需要维护的冲煮头。
- The necessary tools and replacement parts are at this station.
需要的工具及更换零件备妥在本工作站。
- Use the provided tools to:
使用提供的工具进行下列事项:
 - Remove the group screen and gasket.
 - Clean the group.
 - Install a new group screen and gasket.
 - Test the brew group to check for leaks.

拆除冲煮头分水网及垫圈。

清洁冲煮头。

安装新的冲煮头分水网及垫圈。

测试冲煮头是否漏水。

Skill B - Rebuild an espresso machine steam wand joint

技能 B-重新组装一只义式咖啡机的蒸气棒接头

- The trainer has indicated which steam wand is to be serviced and which wear parts are to be replaced.
讲师已经指定需要维护的蒸气棒以及需要更换的耗材。
- The necessary tools and replacement parts are at this station.
需要的工具及更换零件备妥在本工作站。
- Use the provided tools to
使用提供的工具进行下列事项:
 - Remove the steam wand.
 - Remove the wear parts which require replacement.
 - Clean the steam wand.
 - Install new wear parts, as indicated, and reassemble the steam wand.
 - Test the steam wand to check for leaks.

拆除蒸气棒。

拆除需要更换的耗材。

清洁蒸气棒。

根据指示安装新的耗材并重新组装蒸气棒。

测试蒸气棒是否泄漏。

Skill C - Complete an espresso machine inspection

技能 C-完成义式咖啡机的检查

- This section of the Practical Exam Learner Worksheet includes a Preventive Maintenance Inspection Checklist for this espresso machine
本章节的实务测验学员工作表包含一份这台义式咖啡机的预防性维护检查核对表。
- Inspect all of the items indicated and record your observations on your Learner Worksheet
检查所有指定的对象并将观察结果记录在你的学员工作表上。

Scoring 计分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

将得出的小计总分填入实务测验学员工作表的对应表格

Skill 4A - Replace an espresso machine group screen and gasket

技能 4A-更换一组义式咖啡机的冲煮头分水网及垫圈

Task 任务	Points Possible 可能得分
Learner properly removes group screen and screw	2

学员正确地拆除冲煮头分水网及螺丝。	
Learner properly removes group gasket without damaging brewing group 学员正确地拆除垫圈且未损坏冲煮头。	3
Learner cleans group gasket channel and group 学员将垫圈沟槽及冲煮头清洗干净。	2
Learner orients group gasket correctly when installing 学员以正确的一面定位垫圈。	2
Learner properly installs new group gasket 学员正确地安装新垫圈。	2
Learner installs group screen and screw correctly, using appropriate screw torque 学员正确地安装冲煮头分水网及螺丝，并使用适当的力道拧紧螺丝。	2
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1
Total 总分	14

Skill 4B - Rebuild an espresso machine steam wand joint

技能 4B-重新组装一只义式咖啡机的蒸气棒接头

Task 任务	Points Possible 可能得分
Learner properly removes steam wand from steam valve 学员正确地 从蒸气阀上拆下蒸气棒。	2
Learner properly removes wear parts as directed without damage to equipment 学员根据引导正确地拆除磨损零件且未对设备造成损坏。	2
Learner cleans steam wand joint area and valve as directed 学员根据引导将蒸气棒接头区块及蒸汽阀清洗干净。	2
Learner properly installs replacement parts, applying lubricant where applicable 学员正确地安装更换零件，并在适当的地方添加润滑剂。	3
Learner properly reassembles steam wand to steam valve 学员正确地将蒸气棒重新安装回蒸气阀。	2
Steam wand joint does not leak when tested 确认测试时蒸气棒接头不会泄漏。	2

Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1
Total 总分	14

Skill 4C - Complete an espresso machine inspection**技能 C-完成义式咖啡机的检查**

Task 任务	Points Possible 可能得分
Learner correctly records steam boiler pressure on Learner Worksheet 学员正确地在学员工作表上记录蒸气锅炉压力。	1
Learner observes pump pressure and records on Learner Worksheet 学员观测泵浦压力并记录在学员工作表上。	2
Learner inspects safety valve and records operating state on Learner Worksheet 学员检查安全阀并将运作状况记录在学员工作表上。	1
Learner tests anti-vacuum valve and records operating state on Learner Worksheet 学员测试真空排除阀并将运作状况记录在学员工作表上。	2
Learner inspects expansion valve and records operating state on Learner Worksheet 学员检查膨胀阀并将运作状况记录在学员工作表上。	1
Learner tests autofill system and steam boiler water level and records observations on Learner Worksheet 学员测试自动补水系统及蒸气锅炉水位并将观测结果记录在学员工作表上。	2
Learner tests drain function and records operating state on Learner Worksheet 学员测试排水功能并将运作状况记录在学员工作表上。	2
Learner inspects machine for water leaks and records status on Learner Worksheet 学员检查机器是否漏水并将状况记录在学员工作表上。	2
Learner inspects machine for steam leaks and records status on Learner Worksheet 学员检查机器的蒸气是否会泄漏并将状况记录在学员工作表上。	2
Total 总分	15